EXPERIMENT-3

NAME: SANTHOSH KUMAR N ROLL NO:240701807

1.COMMAND LINE INTERFACE:

PYTHON CODE:

```
import os
import sys

def rename_file(old_name, new_name):
    try:
        os.rename(old_name, new_name)
        print(f"File renamed from {old_name} to {new_name}")
    except FileNotFoundError:
        print(f"Error: {old_name} not found.")
    except Exception as e:
        print(f"An error occurred: {e}")

if __name__ == "__main__":
    if len(sys.argv) != 3:
        print("Usage: python rename_file_cli.py <old_filename> <new_filename>")
    else:
        rename_file(sys.argv[1], sys.argv[2])
```

OUTPUT :

```
C:\Users\Santhosh\240701807>notepad renamefile.py

C:\Users\Santhosh\240701807>python renamefile.py file1.txt file2.txt
File renamed from file1.txt to file2.txt

C:\Users\Santhosh\240701807>
```

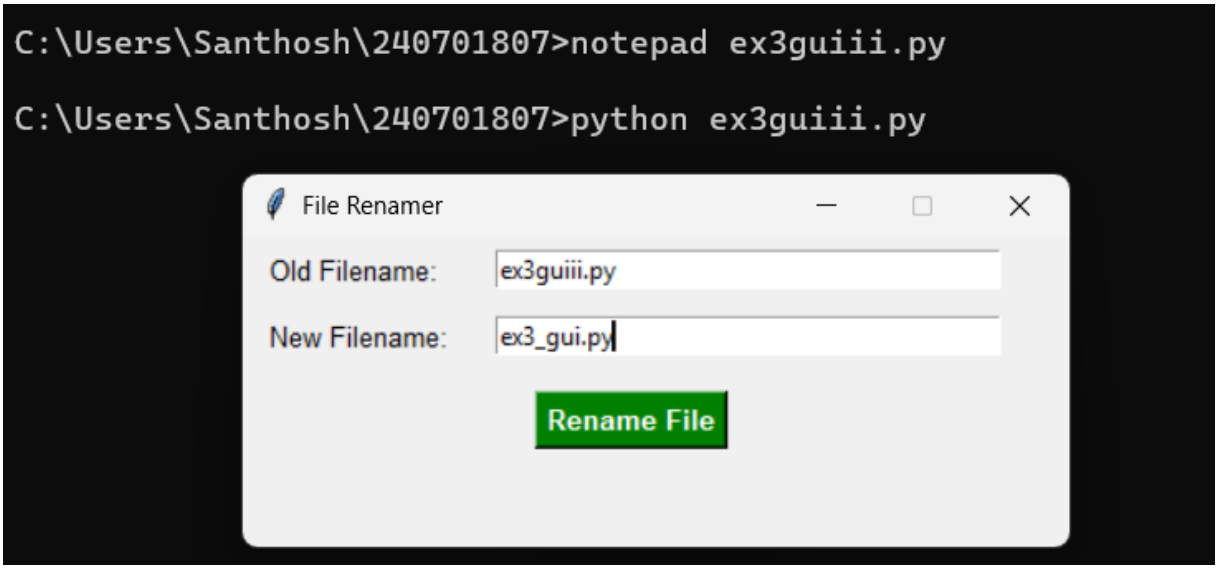
2.GRAPHICAL USER INTERFACE:

PYTHON CODE:

```
import tkinter as tk from tkinter
import messagebox import os

# Function to rename the file def
rename_file():
   old_name = old_filename_entry.get().strip()
new_name = new_filename_entry.get().strip()

   # Check if input fields are empty
if not old_name or not new_name:
      messagebox.showwarning("Warning", "Please enter both filenames!")
      return

   # Check if the old file exists     if
not os.path.exists(old_name):
      messagebox.showerror("Error", f"File '{old_name}' not found.")
      return

   # Check if the new file already exists
if os.path.exists(new_name):
      overwrite = messagebox.askyesno("Warning", f"'{new_name}' already exists. Overwrite?")
      if not overwrite:
         return

   try:
      os.rename(old_name, new_name)        messagebox.showinfo("Success", f"File
renamed from '{old_name}' to '{new_name}'")     except Exception as e:
```

```
    messagebox.showerror("Error", f"An error occurred: {e}")

# Create main window
root = tk.Tk()
root.title("File Renamer")
root.geometry("400x150"
)
root.resizable(False, False)  # Fixed window size

# Labels
tk.Label(root, text="📂 Old Filename:", font=("Arial", 10)).grid(row=0, column=0, padx=10, pady=5, sticky="w")
tk.Label(root, text="📝 New Filename:", font=("Arial", 10)).grid(row=1, column=0, padx=10, pady=5, sticky="w")

# Entry fields old_filename_entry = tk.Entry(root, width=40)
old_filename_entry.grid(row=0, column=1, padx=10, pady=5)

new_filename_entry = tk.Entry(root, width=40) new_filename_entry.grid(row=1,
column=1, padx=10, pady=5)

# Styled Button
rename_button = tk.Button(root, text="✅ Rename File", bg="green", fg="white", font=("Arial", 10, "bold"), command=rename_file) rename_button.grid(row=2,
column=0, columnspan=2, pady=10)

# Run the GUI event loop
root.mainloop()
```

OUTPUT :



3.VOICE USER INTERFACE:

<u>PYTHON CODE:</u>

```
import speech_recognition as sr
import os

def rename_file_from_voice_command(command):
    try:
        words = command.lower().split(" ")
   if "rename" in words and "to" in words:
rename_index = words.index("rename")
to_index = words.index("to")

            # Extract old and new filenames
old_name = words[rename_index + 1]
new_name = words[to_index + 1]
```

```
        # Check if file exists            if
not os.path.exists(old_name):
            print(f"Error: File '{old_name}' not found.")
            return

        # Rename file
        os.rename(old_name, new_name)           print(f"☑ File
renamed from '{old_name}' to '{new_name}'")

else:
        print("Invalid command format. Say: 'Rename oldfile.txt to newfile.txt'")

    except Exception as e:
print(f"Error: {e}")

def listen_for_command():
recognizer = sr.Recognizer()     mic
= sr.Microphone()

    print("🎤 Listening for command to rename a file...")
with mic as source:
        recognizer.adjust_for_ambient_noise(source)
audio = recognizer.listen(source)

try:
        command = recognizer.recognize_google(audio)
print(f"🎙 Command received: {command}")
rename_file_from_voice_command(command)     except
sr.UnknownValueError:
        print("❌ Sorry, I couldn't understand the command.")
except sr.RequestError as e:
```

```
    print(f"⚠ Could not request results from Google Speech Recognition service; {e}")


if __name__ == "__main__":
listen_for_command()
```

OUTPUT :

```
C:\Users\Santhosh\240701807>python voice.py
🎤 Listening for command to rename a file...
❌ Sorry, I couldn't understand the command.

C:\Users\Santhosh\240701807>python voice.py
🎤 Listening for command to rename a file...
🎙 Command received: rename
Invalid command format. Say: 'Rename oldfile.txt to newfile.txt'
```

```
C:\Users\Santhosh\240701807>python voice.py
🎤 Listening for command to rename a file...
🎙 Command received: rename sample.txt to new.txt
✅ File renamed from 'sample.txt' to 'new.txt'
```

4.USER SATISFACTION COMPARISON:

<u>PYTHON CODE:</u>

```
def survey():
  print("Rate your satisfaction with the following interfaces (1-5):")


  # Get user input for each interface
  try:
    cli_satisfaction = int(input("CLI (1-5): "))
gui_satisfaction = int(input("GUI (1-5): "))
vui_satisfaction = int(input("VUI (1-5): "))


    # Ensure valid ratings        if not (1 <= cli_satisfaction <= 5 and 1 <= gui_satisfaction <= 5 and
1 <= vui_satisfaction <= 5):
```

```
            print("Please enter ratings between 1 and 5 only.")
            return

        # Display the ratings
print("\nYour satisfaction ratings:")
print(f"CLI: {cli_satisfaction}")
print(f"GUI: {gui_satisfaction}")
print(f"VUI: {vui_satisfaction}")


        # Calculate the average satisfaction        avg_satisfaction =
(cli_satisfaction + gui_satisfaction + vui_satisfaction) / 3        print(f"\nAverage
Satisfaction Score: {avg_satisfaction:.2f}")


    except ValueError:
        print("Invalid input! Please enter numbers between 1 and 5.")
# Run the survey function if
__name__ == "__main__":
    survey()
```

OUTPUT :

```
C:\Users\Santhosh\240701807>notepad ratings.py

C:\Users\Santhosh\240701807>python ratings.py
Rate your satisfaction with the following interfaces (1-5):
CLI (1-5): 2
GUI (1-5): 5
VUI (1-5): 3

Your satisfaction ratings:
CLI: 2
GUI: 5
VUI: 3

Average Satisfaction Score: 3.33
```